

POKHARA UNIVERSITY

Level: Bachelor

Semester: Spring

Year: 2024

Programme: BBA/BBA-BI/BBA-TT/BCIS

Full Marks: 100

Course: Data Analysis and Modeling

Pass Marks: 45

Time: 3 hrs.

Candidates are required to answer in their own words as far as practicable. The figures in the margin indicate full marks.

Section "A"

Very Short Answer Questions

Attempt all the questions. [10×2]

1. Mention any two basic differences between correlation and regression.
2. Interpret the slope of the regression model $\hat{Y} = 35 - 10X$. What is the estimated value of Y when X=1.5.
3. Why Fisher's index number is called ideal index number?
4. What do you understand by balanced and unbalanced transportation problems?
5. Find the active and inactive constraints in the following LPP;
 $Maximize Z = 2x + 4y$
 $Subject to x + y \leq 14, 3x + 2y \geq 30, 2x + y \leq 18, x \geq 0, y \geq 0.$
 $where max z = 36 at x = 6 and y = 6$
6. What do you mean by time series analysis?
7. From the given information calculate forecast value for the month April using Exponential smoothing constant 0.1 and forecast for January was 195 units.

Month	Demand	Forecast
January	200 units	195 units
February	110 units	?
March	300 units	?
April	-	?

8. The correlation coefficient between two variables Y and X is 0.85. Calculate the value of coefficient of determination and interpret its meaning.
9. Define regression analysis.
10. Draw network diagram for the following data:

Activity	A	B	C	D	E	F	G
Predecessors	-	-	A	B	B	CD	EF

Section "B"

Descriptive Answer Questions

Attempt **any six** questions. [6×10]

11. A farm administers a test to sales trainees before they go into the field. The management of the farm is interested in determining the relationship between the test scores and the sales made by the trainees at the end of one year in the field. The following data were collected for 10 sales personnel who have been in the field:

Descriptive Answer Questions

Attempt **any six** questions. [6×10]

11. A farm administers a test to sales trainees before they go into the field. The management of the farm is interested in determining the relationship between the test scores and the sales made by the trainees at the end of one year in the field. The following data were collected for 10 sales personnel who have been in the field:



1 of 4

Sales Person Number	Test Score(X)	Number of units sold(Y)
1	2.6	95
2	3.7	140
3	2.4	85
4	4.5	180
5	2.6	100
6	5.0	195
7	2.8	115
8	3.0	136
9	4.0	175
10	3.4	150

Calculation shows that:

$$\sum X = 34 \quad \sum Y = 1371 \quad \sum X^2 = 122.62 \quad \sum Y^2 = 201121 \quad \sum XY = 4954$$

- Find the correlation coefficient between the test score and number of units sold, examine if this linear relationship is significant at the 5% level of significance.
- Compute the 90% prediction interval for the number of units sold of salesperson 9.
- Find coefficient of determination and interpret its meaning.

12. A certain fast food has data on the number of boxes of cereal it has sold in each of the last 7 years.

Year	1989	1990	1991	1992	1993	1994	1995
Boxes	210	194	226	282	304	240	250

- Find the linear estimating equation that best describes these data.
 - Calculate the percentage of trend for these data.
 - Calculate the relative cyclic residual for these data.
 - In which year does the biggest fluctuation from the trend occur, under each measure of cyclic variation? Is this year the same for both methods? Explain.
13. a. From the following data, find the appropriate price index number with due justification.

Commodity	2015		2020	
	Price	Quantity	Price	Quantity
A	15	225	18	240
B	20	275	25	270
C	25	250	30	300

- Define index number. Describe about the uses of index numbers.
14. A carpenter has 90, 80 and 50 meters of teak, plywood and rosewood. The product A requires 2, 1 and 1 unit and product B requires 1, 2 and 1 units of teak, plywood and rosewood respectively. If product A and B have profits of Rs.48 and Rs.40 per unit respectively. Then how much of it product should be produced to make maximum profit. Formulate the problem and find graphical solution of it.



15. Solve the following transportation problem for minimum cost by taking initial feasible solution by Vogel's Approximation Method. The entries in the matrix indicate the cost in rupees of transporting a unit from a particular source to a particular destination.

Origin	Destination				Availability
	D ₁	D ₂	D ₃	D ₄	
O ₁	10	8	11	7	20
O ₂	9	12	14	6	40
O ₃	8	9	12	10	35
Requirement	16	18	31	30	95

16. a. The following table shows the number of visitors who tourist area over a 10-year period.

Year	1998	1999	2000	2001	2002	2003	2004	2005	2006	2007
No. of visitors	28	27	33	25	34	33	35	30	33	35

- Forecast the number of visitors in 2008 by using Naïve model. Use the first 6 period as a warm up sample.
 - Also compute the MSE and MAD for the forecasting sample.
- b. There are four machine on which to do three jobs. The cost (in Rs.) of each job on each machine is given in the following table.

Job	Machine			
	W	X	Y	Z
A	18	24	28	32
B	8	13	17	19
C	10	15	11	22

Assign the job to each machine to minimize costs.

17. A project consist the following activities.

Activity	Predecessors	Time estimates		
		Optimistic	Most likely	pessimistic
A	-	3	4	5
B	-	2	2	2
C	A	1	3	5
D	B,C	0	4	14
E	D	1	2	3
F	A	2	9	10

- Draw a project network and determine the critical path.
- What is the probability that the project will be completed within 12 weeks?
- What deadline is consistent with a 0.95 probability of on time completion?

Section "C"
Case Analysis

18. Read the case situation given below and answer the questions that follow:

[20]

A manager selected a representative sample of 24 monthly customer bills taken from several recent heating seasons. The manager considers kilowatt hours per month (Y) as a liner function of square feet heated space (X_1), an index of roof insulation quality (X_2), PRESENCE/ABSENCE of insulated windows (X_3), mean temperature (X_4) and heat pump/electric forced air (X_5). **The SPSS output** is as following:

	Unstandarized Coefficients(β)	Std.error	T	Sig.
Intercept	6356.07	838.701	7.58	0.0000
X_1	0.56038	0.15811	3.54	0.0023
X_2	-31.2077	8.95905	-3.48	0.0027
X_3	-327.503	149.169		
X_4	-113.895	16.2604	-7.00	0.0000
X_5	-621.485	147.828	-4.20	0.0005

ANOVA TABLE

Sources	Degree of freedom (d.f.)	Sum of square (SS)	Mean square (MS)	F
Regression	5			
Residual		2166000		
Total	23	14370000		

- Fit a multiple regression equation.
- Test which of the independent variable makes significant contribution to the model.
- Compute the standard error of the estimate (S_e).
- Compute the r^2 and interpret its meaning.
- Obtain confidence interval estimate of coefficient of X_3 .
- Given that $X_1 = 129$, $X_2 = 18$, $X_3 = 5$, $X_4 = 3$, $X_5 = 1129$, estimate the value of Y.
- Set up the null and alternative hypothesis, carry out F-test and interpret your result.